

9

Circles



The Ferris wheel is one of the attractions that most children seek out when visiting amusement parks. For creating such structures, circles are preferred as it provide equal support to handle the desired load in each of the components. These components maintain the overall balance of the wheel.



What's Inside

Topic(s)

- ✓ Topic-wise Notes
- ✓ Related NCERT Qs
- ✓ Caution Points
- ✓ Case-based Examples

Questions

- ✓ New Pattern Qs
- ✓ All Objective + Subjective Qs
- ✓ Diksha + Exemplar Qs
- ✓ Detailed Explanation

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TOPIC 1

ANGLE SUBTENDED BY A CHORD AT A POINT

Learning Objectives

- Students will be learning that the angle subtended by the equal chords of a circle at the centre are equal.

Learning Outcomes

- Students will be able to understand the theorems and apply them.

Real Life Application

If you have been to Delhi for an educational tour, you must have seen the Parliament house. In the aerial view of the Parliament House of India circle shaped structures instead of conventional ones are observed. You must have wondered what goes into creating buildings or structures with circular shapes.



So, here, the architects and engineers applied the concept of circles in their design so that the building has a circular shape and is also practical and strong enough.

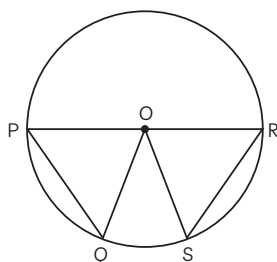
To make such circular structures, concepts of arcs, chords and angles are used. This is how angle subtended by a chord applies in real life. Let's study angle subtended by a chord at a point in detail.

As we know, the chord of circle is the line segment that joins two points on the circumference of the circle. If we draw two or more equal chords of a circle and determine the angles subtended by them at the centre, we will observe that the size of the chord and the angle subtended by it at the centre has a relationship.

Theorem-1

Equal chords of a circle subtend equal angles at the centre.

Given: A circle with centre O.



PQ and RS are equal chords

i.e., $PQ = RS$

To Prove: $\angle POQ = \angle ROS$

Proof: In $\triangle POQ$ and $\triangle ROS$

$$OP = OR$$

[Radius]

$$OQ = OS$$

[Radius]

$$PQ = RS$$

[Given]

$$\triangle POQ \cong \triangle ROS$$

[By SSS Congruence Rule]

$$\therefore \angle POQ = \angle ROS$$

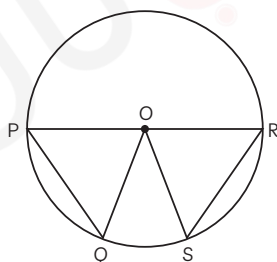
[By CPCT]

Hence, proved.

Theorem-2

If the angles subtended by the chords of a circle at the centre are equal, then the chords are equal.

Given: A circle with centre O.



PQ and RS are chords.

$$\angle POQ = \angle ROS$$

To prove: $PQ = RS$

Proof: In $\triangle POQ$ and $\triangle ROS$

$$OP = OR$$

[Radius]

$$OQ = OS$$

[Radius]

$$\angle POQ = \angle ROS$$

[Given]

Therefore, $\triangle POQ \cong \triangle ROS$ [By SAS Congruence Rule]

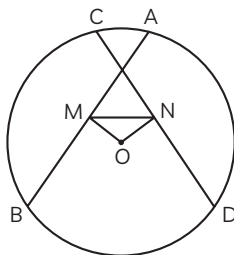
$$\text{So, } PQ = RS$$

[By CPCT]

Hence, proved.



Example 1. Show that $\angle OMN = \angle ONM$, if AB and CD are two equal chords.



Ans. We know that equal chords of a circle are equidistant from the centre.

$$AB = CD$$

$$\Rightarrow OM = ON$$

In $\triangle OMN$,

$$OM = ON$$

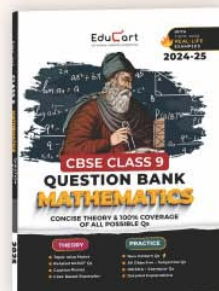
$$\angle OMN = \angle ONM$$

[Angles opposite to equal sides]



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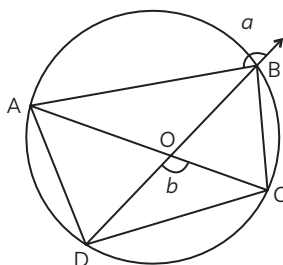


OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. Given below is a circle with centre O. Which of the following represents the measure of $\angle BCD$?



(a) $180^\circ + (a - \frac{b}{2})$

(b) $180^\circ - (a + \frac{b}{2})$

(c) $90^\circ - (a - \frac{b}{2})$

(d) $180^\circ - (a - \frac{b}{2})$

[Diksha]

Ans. (d) $180^\circ - (a - \frac{b}{2})$

Explanation: From figure

$$\angle AOB = \angle DOC$$

[Vertically opposite angles]

Here, $\angle AOB = \angle DOC = b$

Now, $\angle ACB = \frac{1}{2} \angle AOB$

[Angle made by the same segment on the circumference is half of that made on the centre]

So, $\angle ACB = \frac{b}{2}$

Now, DB is a straight line

$$\angle ABO + a = 180^\circ$$

[Linear pair]

$$\angle ABO = 180^\circ - a$$

$$\angle ACD = \angle ABO = 180^\circ - a$$

[Chord AD is Common]

Now, $\angle BCD = \angle ACB + \angle ACD$

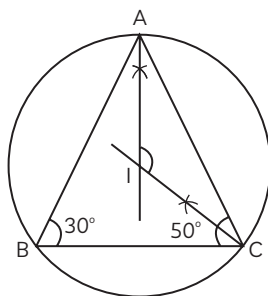
$$= \frac{b}{2} + 180^\circ - a$$

$$\angle BCD = 180^\circ - a + \frac{b}{2}$$

$$= 180^\circ - (a - \frac{b}{2})$$

2. A triangle ABC is inscribed in a circle with $\angle B = 30^\circ$ and $\angle C = 50^\circ$. The angle bisectors of $\angle A$ and $\angle C$ meet at I. the measure of $\angle AIC$ is:





(a) 105°

(b) 100°

(c) 75°

(d) 120°

Ans.(a) 105°

Explanation: In $\triangle ABC$,

$$\angle A + \angle B + \angle C = 180^\circ$$

[Angle sum property]

$$\angle A + 30^\circ + 50^\circ = 180^\circ$$

$$\angle A = 180^\circ - 30^\circ - 50^\circ = 100^\circ$$

Since, we know that angle bisector divides the angle in equal parts.

Therefore, $\angle ACI = \frac{\angle C}{2} = \frac{50^\circ}{2} = 25^\circ$

$$\angle CAI = \frac{100^\circ}{2} = 50^\circ$$

In $\triangle AIC$,

$$\angle ACI + \angle CAI + \angle AIC = 180^\circ$$

$$25^\circ + 50^\circ + \angle AIC = 180^\circ$$

$$\angle AIC = 180^\circ - 25^\circ - 50^\circ$$

$$\angle AIC = 105^\circ$$

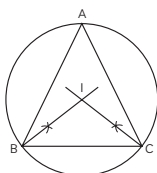
⚠ Caution

→ Students should remember that the incentre of the triangle can also be found by formula (given below)

$$\angle BIC = 90^\circ + \frac{\angle A}{2}$$

$$\angle AIC = 90^\circ + \frac{\angle B}{2}$$

$$\angle AIB = 90^\circ + \frac{\angle C}{2}$$



True and False

3. An equilateral triangle can be inscribed in a semicircle with its one side as the diameter of the semicircle.

Ans. False

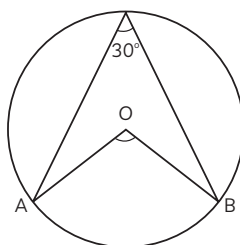
Explanation: It is not possible to inscribe an equilateral triangle in a semicircle with one side as diameter of the semicircle because all the

sides of the equilateral triangle are equal and form congruent angles, also we know that the angle in a semicircle is 90° .

Hence, the answer is false.

Fill in the Blanks

4. In a circle with centre O, if the angle subtended by minor arc AB at the boundary of the circle is 30° , then the measure of $\angle AOB$ will be



Ans. 60°

Explanation: As we know the angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

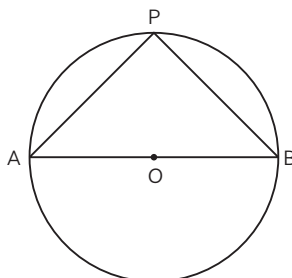
Assertion and Reason (A-R)

In these questions, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.

5. Assertion (A): AB is the diameter of a circle with centre O. P is a point on the circumference of the circle. Such that $PA = PB$, then $\angle PAB = 45^\circ$



Reason (R): Angle in a semicircle is a right angle.



Ans.(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

Explanation:

In $\triangle APB$,

$$PA = PB$$

[Given]

$$\text{So, } \angle PAB = \angle PBA$$

[Angle of an isosceles triangle]

$$= x \text{ (say)}$$

Now, angle in a semicircle is a right angle.

$$\angle APB = 90^\circ$$

[AB is diameter]

$$\text{So, } \angle PAB + \angle PBA + \angle APB = 180^\circ$$

$$x + x + 90^\circ = 180^\circ$$

$$x = \frac{90^\circ}{2} = 45^\circ$$

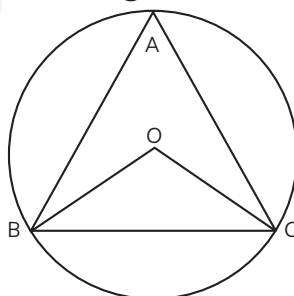
Hence, both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

CASE BASED Questions (CBQs)

[4 & 5 marks]

Read the following passages and answer the questions that follow:

6. Three STD booths are placed at A, B and C in the figure and these are operated by handicapped persons. These three booth are equidistant from each other as shown in the figure.



(A) Find the measure of $\angle BAC$.

(B) Find the measure of $\angle BOC$.

(C) If $AB = 8$ cm, find the value of $(BC + CA)$.

[British Council 2022]

Ans.(A) In the figure, $AB = BC = CA$.

$\therefore \triangle ABC$ is an equilateral triangle.

$$\Rightarrow \angle ABC = \angle ACB = \angle BAC = 60^\circ$$

(B) It is given that OB is the bisector of $\angle ABC$.

$$\therefore \angle ABO = \angle OBC = \frac{1}{2} \angle ABC$$

$$= \frac{1}{2} \times 60^\circ = 30^\circ \quad \dots (i)$$

Also, OC is the bisector of $\angle ACB$.

$$\therefore \angle ACO = \angle OCB = \frac{1}{2} \angle ACB$$

$$= \frac{1}{2} \times 60^\circ = 30^\circ \quad \dots (ii)$$

In $\triangle OBC$,

$$\angle OBC + \angle OCB + \angle BOC = 180^\circ$$

[Angle sum property]

$$\Rightarrow 30^\circ + 30^\circ + \angle BOC = 180^\circ$$

$$\Rightarrow \angle BOC = 180^\circ - 60^\circ = 120^\circ$$

(C) $\triangle ABC$ is an equilateral triangle.

$$AB = BC = CA$$

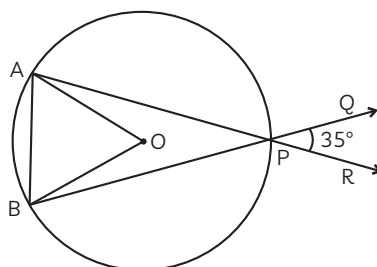
$$BC + CA = 8 + 8$$

$$\text{Therefore } BC + CA = 16 \text{ cm.}$$

VERY SHORT ANSWER Type Questions (VSA)

[1 mark]

7. In the following figure, find $\angle AOB$.



[Diksha]

Ans. $\angle APB = \angle RPQ = 35^\circ$



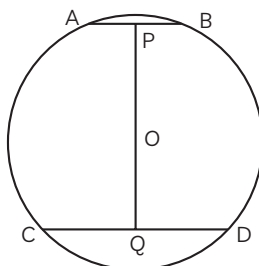
Now, $\angle AOB$ and $\angle APB$ are angles subtended by an arc AB at centre and at the remaining part of the circle.

$$\therefore \angle AOB = 2\angle APB = 2 \times 35^\circ = 70^\circ$$

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

8. If the radius of a circle is 5 cm, $AB = 6$ cm and $CD = 8$ cm, then determine PQ.



[British Council 2022]

Ans. Since, the perpendicular from the centre of a circle to a chord bisects the chord.

Therefore, P and Q are mid-points of AB and CD respectively.

$$AP = PB = \frac{1}{2} AB = 3 \text{ cm}$$

$$\text{and } CQ = QD = \frac{1}{2} CD = 4 \text{ cm}$$

In right-angled $\triangle OAP$ and $\triangle OCQ$ by using Pythagoras theorem

$$OA^2 = OP^2 + AP^2 \text{ and } OC^2 = OQ^2 + CQ^2$$

$$OP^2 = 5^2 - 3^2 \quad OQ^2 = 5^2 - 4^2$$

$$OP^2 = 16 \quad \quad \quad = 9$$

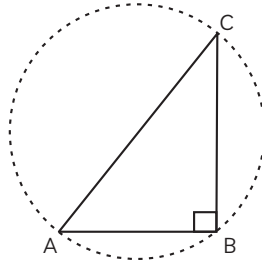
$$\Rightarrow OP = 4 \quad \text{and } OQ = 3$$

$$PQ = OP + OQ = (4 + 3) = 7 \text{ cm}$$

SHORT ANSWER Type-II Questions (SA-II)

[3 marks]

9. If in a $\triangle ABC$, $AB = 12$ cm, $BC = 16$ cm and $AB \perp BC$, then find the radius of the circle passing through A, B and C.



[Delhi Gov. QB 2022]

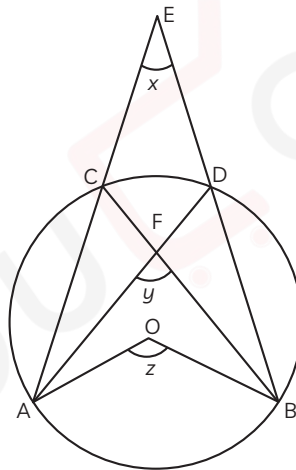
Ans.Given: A, B and C are points such that $AB = 12$ cm and $BC = 16$ cm and $BC \perp AB$.

$BC \perp AB$ i.e., $\angle ABC = 90^\circ$ and $\triangle ABC$ is a right one with AC as hypotenuse.

$\therefore AC = \sqrt{AB^2 + BC^2} = \sqrt{12^2 + 16^2}$ cm = 20 cm. So, the circle passing through A, B and C will have its diameter as AC.

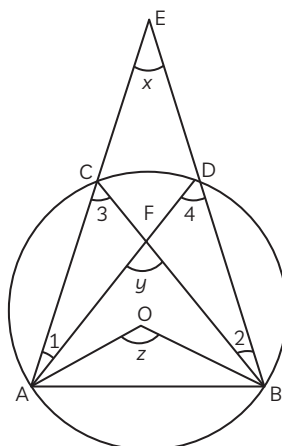
$\therefore$ Its radius = $\frac{1}{2} \times 20$ cm = 10 cm.

10. In the given figure, O is the centre of a circle prove that $\angle x + \angle y = \angle z$.



[Delhi Gov. QB 2022]

Ans.In $\triangle ACF$, side $\angle CF$ is produced to B.



$$\therefore \angle y = \angle 1 + \angle 3 \quad \dots(i)$$

[Ext. angle = sum of int. opp. angle]

In $\triangle AED$, side ED is produced to B .

$$\therefore \angle 1 + \angle x = \angle 4 \quad \dots(ii)$$

Add eqs. (i) and (ii), we have

$$\begin{aligned} \angle 1 + \angle x + \angle y \\ = \angle 1 + \angle 3 + \angle 4 \end{aligned}$$

$$\begin{aligned} \Rightarrow \angle x + \angle y &= \angle 3 + \angle 4 \\ &= 2 \angle 3 \end{aligned}$$

[$\because \angle 4 = \angle 3$, angles in the same segment]

$$= \angle z$$

[$\because \angle AOB = 2\angle ACB$]

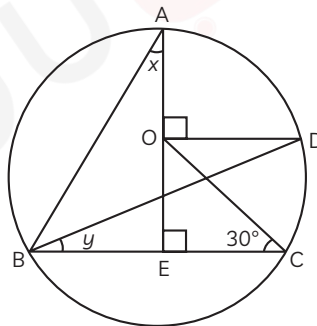
$$\angle x + \angle y = \angle z$$

Hence, proved.

LONG ANSWER Type Questions (LA)

[4 & 5 marks]

11. In figure, O is the centre of the circle, $\angle BCO = 30^\circ$, $AE \perp BC$ and $DO \perp AE$. Find x and y .



[Delhi Gov. QB 2022]

Ans. In the given figure, OD is parallel to BC .

$$\therefore \angle BCO = \angle COD$$

[Alternate interior angles]

$$\Rightarrow \angle COD = 30^\circ \dots (i)$$

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it on the remaining part of the circle.

Here, arc CD subtends $\angle COD$ at the centre and $\angle CBD$ at B on the circle.

$$\therefore \angle COD = 2\angle CBD$$

$$\Rightarrow \angle CBD = \frac{30^\circ}{2} = 15^\circ$$

$$\therefore y = 15^\circ \quad \dots\dots (ii) \quad \text{[From (i)]}$$

Also, arc AD subtends $\angle AOD$ at the centre and $\angle ABD$ at B on the circle.

$$\therefore \angle AOD = 2\angle ABD$$

$$\Rightarrow \angle ABD = \frac{90^\circ}{2} = 45^\circ \dots\dots (iii)$$

In $\triangle ABE$,

$$x + y + \angle ABD + \angle AEB = 180^\circ$$

$$\Rightarrow x + 15^\circ + 45^\circ + 90^\circ = 180^\circ$$

$$\Rightarrow x = 180^\circ - (90^\circ + 15^\circ + 45^\circ)$$

$$\Rightarrow x = 180^\circ - 150^\circ$$

$$\Rightarrow x = 30^\circ \quad \text{[Sum of the angles of a triangle]}$$

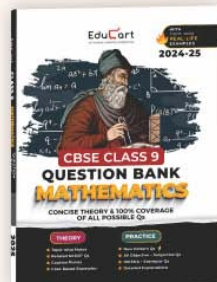
[From eqs. (ii) and (iii)]

Hence, $x = 30^\circ$ and $y = 15^\circ$.



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